

Measurement System for the Characterization of the Human Body as a Communication Channel at Low Frequency

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Abstract—Electronic data transfer by capacitive and galvanic coupling through the human body has been proposed by research and industry as a novel but highly promising technology for ultra low power wireless body LANs. Investigation on the most challenging questions considering data communication becomes enabled with a highly versatile measurement system for frequencies in the range of 10 kHz to 1 MHz. The human body is characterized as a transmission medium for electrical current by means of measurements and is investigated as communication channel for biomedical parameter monitoring by using different modulation schemes at low frequency.

Excellent transmission was achieved on the thorax while the attenuation increases along the extremities.

The injected current is 10 times below the maximum allowed contact current and more than 25 times below nerve stimulation. The new technology has shown its feasibility in clinical trials.

I. INTRODUCTION

Driven by the vision of transmitting data over and through the human body several attempts have been made to model the human body as electrical channel. With *Personal Area Network (PAN)* [1] a first data communication was realized. Different electrical coupling methods have shown the potential of data communication through the human body [2][3][4][5]. A further principle using electro-optic sensors opened the field for signals up to several MHz [6]-[7].

All these systems used a fixed carrier frequency and showed the feasibility of electrical signal propagation through the human body.

In this paper, a versatile measurement system (fig.1) for the investigation of signals over a wide range from 10 kHz to 1 MHz is proposed. Attenuation and phase response are acquired and suitable modulation types evaluated. Based on the measurements of the

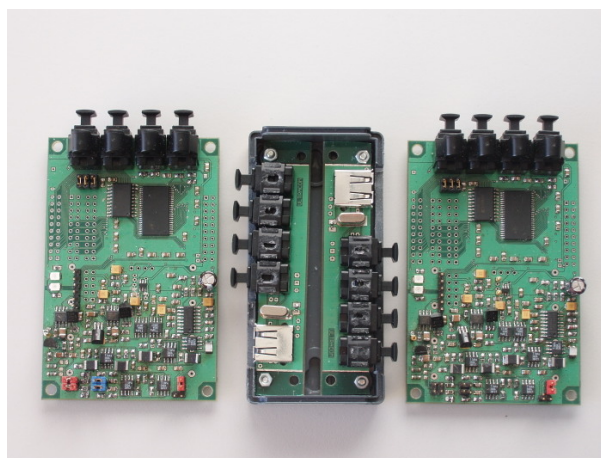


Fig. 1. Transmitter, interface and receiver PCB

electrical behavior of the human body and the observed variations between different paths, the range of possible applications using the human body as a communication medium are identified.

II. MEASUREMENT METHOD

A. Concept of Transmission

A data transfer is established between transmitter and receiver unit by coupling signal currents galvanically into the human body. The transmitter establishes a modulated electrical field which is sensed by the receiver as shown in fig. 2. The current induced by the transmitter has a peak amplitude of 2 mA. The signal is applied differentially over two transmitter electrodes and received by the two differential receiver electrodes.

B. On Body Measurements

In order to assess the characteristics of different paths (extremities, thorax, joints), the following test protocol has been defined:

- Assessment of signal attenuation over frequency by sweeping a sinusoidal signal
- Estimation of phase shift (group delay relative to a fixed reference frequency)

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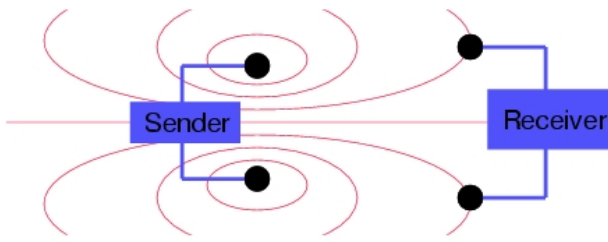


Fig. 2. Concept of transmission using the body as electrical transmission channel

- Measurement of coupling impedance of the channel (signal current of 2 mA is induced and the input voltage is measured)
- Reliability of data transmission depending on transmission schemes (FSK and BPSK modulation)

III. SYSTEM ARCHITECTURE

The low frequency measurement system is composed of two sensor units (transmitter and receiver) and an interface board (see fig. 1). This board offers a serial optical connection to each of the two sensor units and an USB interface to a standard computer.

Basically, the two handheld sensor units allow to send and receive modulated electrical signals. A Xilinx SPARTAN IIE FPGA constitutes the central unit and interfaces between analog front end and digital communication interfaces.

Transmitter and receiver unit both contain the identical analog front end by reason of recording data. The signals generated by the FPGA are fed through a 12 bit digital-to-analog converter into a differential current output with a peak amplitude of 2 mA. The current is band-pass filtered before being coupled into the body. The transmitted signals are differentially picked up by the receiver, band-pass filtered and amplified up to 72 dB. Finally, the signals are digitized by a 14 bit analog-to-digital converter at a sampling rate of 8 MSPS.

The electrical signals are coupled into the human tissue through two standard electrodes being attached to the skin fig. 3. For measurement purposes, transmitted signals are recorded on both transmitter and receiver units.

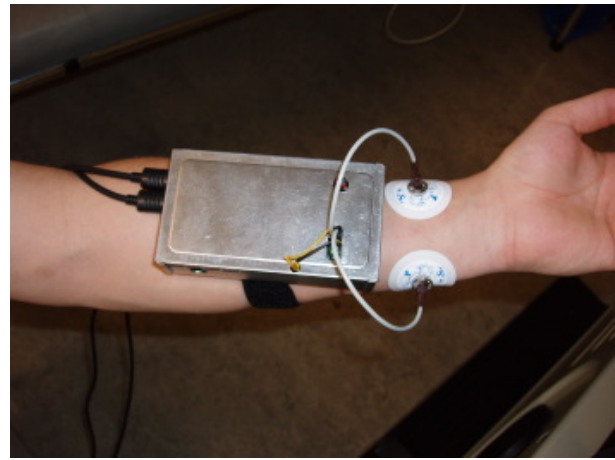


Fig. 3. Transmitter configuration for measurements

Configuration and data exchange are provided by a bidirectional serial data link from each sensor to a host system. The data is received, stored into the SRAM in the sensor and transferred over the serial optical link to the host system for further evaluations. The serial optical data stream is converted to USB on the interface board. The basic functionality on the host is implemented in *Perl* using *win32api* libraries.

Both sensor units are battery powered and fully optically isolated. Therefore, the sensor units are entirely electrically decoupled from any power lines and operate far below safety limits.

IV. CLINICAL TRIAL

The feasibility of the technology was evaluated within a clinical trial. A wide variation of biometric and medical parameters and their influence on electrical signal transmission were investigated.

The measurement sensors are attached to the human body as shown in fig. 3. Both transmitter electrodes are attached in a distance of 5 cm while the receiver electrodes were separated by 7 cm or closer if forced by the body geometry.

Since measurements with any patients are restricted to 3 hours it has been necessary to limit the number of different locations for transmission measurements. Fig. 4 shows those locations and table I indicates the combinations of transmitter and receiver points.

The measurements are grouped into the four sessions along the arm, through the thorax, along the leg, and through the entire body.

The index *geometrical* means a wider separation of the receiver electrodes. Both plasters are displaced to

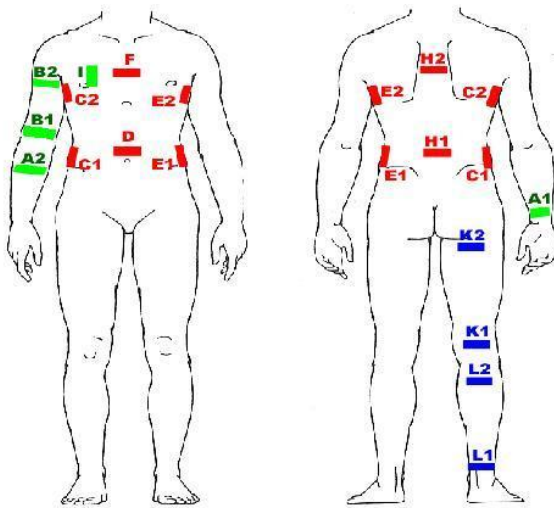


Fig. 4. Measurement point distribution over the entire body

TABLE I
BODY POINTS AND SESSION CONFIGURATIONS

Session	Sender	Receiver
Arm	A1	A2, B1, A2Ergo
	A2	B1, B2
	B1	B2, I
	B2	I
Thorax	C1	E1
	C2	E2
	D	H1
	F	D, Dv, H1, H2
Leg	F	H2EKG, H2Ergo
	L1	L2, K1
	L2	K1, K2
	K1	K2, H1
Path	K2	H1
	F	L1, K2, A1
	H2	Dv
	K2	H1geometrical
	A1, H2, L1, K2	Dgeometrical

diametrical sides of the thorax. Furthermore index v implies a rotation of the two electrodes by 90 degrees.

A. Attenuation results

Measurements on the thorax yield results that are relatively independent of the location and in general show reasonable transmission strength in the range of -50 to -65 dB (fig. 5). The attenuation on arms and legs increases exponentially with distance. So, transmission over large distances gets rather difficult using the low amplitude of maximum 2 mA injected

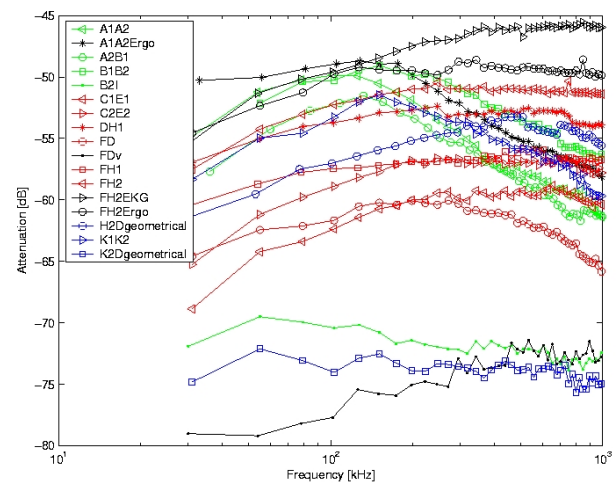


Fig. 5. Attenuation of different body locations indexed with transmitter followed by receiver location

current and 3 V power supply. Especially joints act as blockage and add an attenuation of more than 10 dB.

Fig. 6 shows in detail the average attenuation and standard deviation among all patients with transmitter location $A1$ and receiver $A2$. This configuration represents a short-range communication path along the extremities. The high-pass characteristic below 30 kHz originates from the hardware while the low pass behavior above 200 kHz is a well known characteristic of the human body [2].

While attenuation on the thorax is rather independent of the location, geometrical orientation is highly relevant. 90 degrees of rotation deteriorates the signal detection and results in an attenuation difference of more than 20 dB (see fig. 5 comparing the recorded

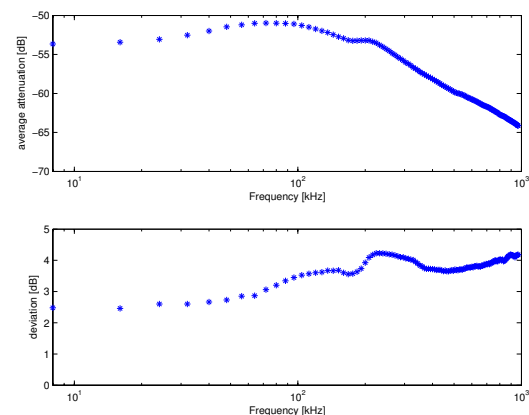


Fig. 6. Attenuation between arm locations $A1$ and $A2$ averaged over all patients

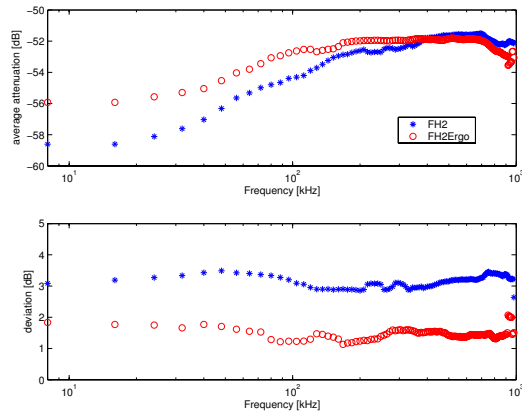


Fig. 7. Attenuation from chest *F* to back *H2* averaged over all patients regular and during ergo meter exposure

signal with index v).

Interestingly, measurements under the impact of body activity on an ergo meter have not shown any degradation. In fact the more activity the better the coupling and the more stable the signal transmission. Fig. 7 shows an average attenuation of -52 dB when applying the signal between chest *F* and back *H2*. The standard deviation is lower than the results obtained on the extremities (fig. 6).

B. Digital modulation results

Frequency-shift keying (FSK) and *binary phase-shift keying (BPSK)*, two common digital modulations for data transmission, have been used for transmitting random bit patterns. Fig. 8 shows the demodulated signals

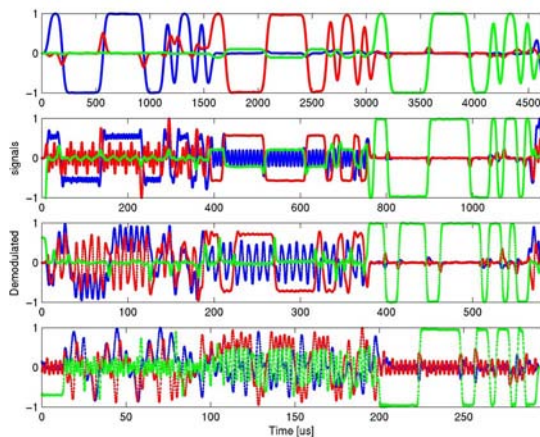


Fig. 8. BPSK data transmission with 8 (top), 64, 128 and 255 kbit/s (bottom) and different carrier frequencies of 80 (blue), 200 (red) and 600 (green) kHz

of BPSK transmission. A data rate up to 255 kbit/s was achieved with a carrier frequency of 600 kHz.

FSK modulation poses higher requirements to the analog part. This data transmission worked down to a frequency shift of 16 kHz and up to a data rate of 128 kbit/s.

V. CONCLUSIONS

The new measurement system offers sophisticated possibilities for signal application and data transmission. High signal to noise ratio is achieved by using a differential pair of electrodes galvanically coupled to the human body.

Different measurement sessions allowed signal applications over a frequency range from 10 k to 1 MHz. Data transfer up to 255 kbit/s was achieved with common digital modulation types.

Clinical trials showed the feasibility of the new technology to transmit data through the human body. With the proposed measurement system dedicated body parts are characterized. High variations of the transmission attenuation has been observed at different locations of the body. Especially the thorax showed excellent transmission characteristics while the extremities and joints lowered the possibility for transmission over larger distance. Improvements have been seen during body activity.

Based on these results the system will be enhanced and miniaturized with the goal of using this method of data transmission in a biomedical system for monitoring vital functions.

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